

A UNIFIED DOWNSTREAM BENCHMARK FOR AN IMMUNE-RECEPTOR SEQUENCE FOUNDATION MODEL

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ABSTRACT

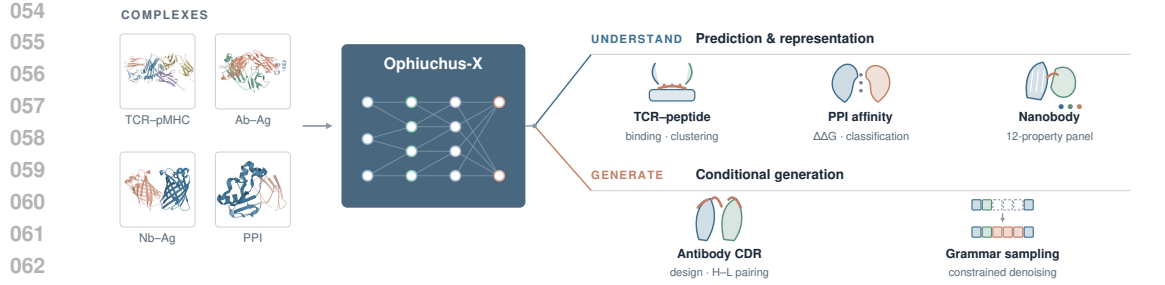
We present a unified downstream benchmark for OPHIUCHUS-X, a grammar-conditioned ESMC→LLaDA diffusion language model for immune-receptor and protein-protein interaction (PPI) sequences. The benchmark spans ten tasks across four families — T-cell receptor (TCR) binding, clustering, representation, and generation; MINT GeneralPPI classification and regression; antibody CDR infilling, light-chain pairing, and FLAb property regression; and the 12-task NbBench nanobody suite — all built from official splits and reproduced against the anchor papers’ own numbers where such numbers exist. A single string-spec interface plugs any checkpoint into every task with post-LLaDA feature extraction. On the retrained integrated 7-layer checkpoint, OPHIUCHUS-X is competitive on antibody CDR-H3 recovery, MINT affinity regression, and several NbBench probes, while trailing specialised baselines on TCR binding/clustering/few-shot, PPI classification, and epitope-conditioned TCR design — reported without inflation.

1 INTRODUCTION

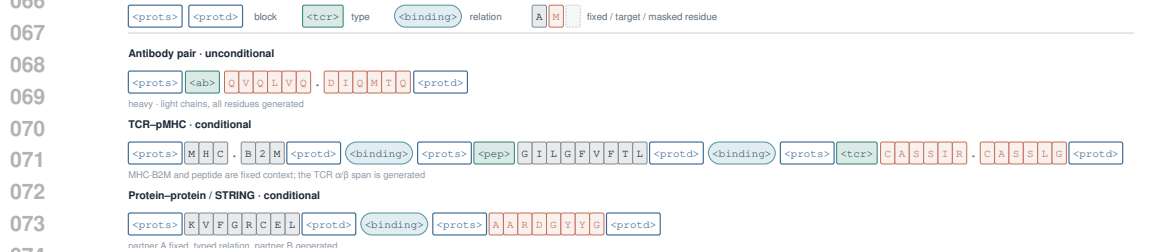
Sequence foundation models are increasingly applied to immune receptors (T-cell receptors, antibodies, nanobodies) and to protein-protein interaction (PPI) modelling. Evaluating such a model fairly requires a broad, reproducible downstream panel that (i) reuses *official* splits and baselines rather than ad-hoc reimplementations, and (ii) reports positioning honestly, including tasks where the model does not win.

This paper documents such a panel for OPHIUCHUS-X, a grammar-conditioned ESMC→LLaDA diffusion language model (Figure 1). Our contributions are:

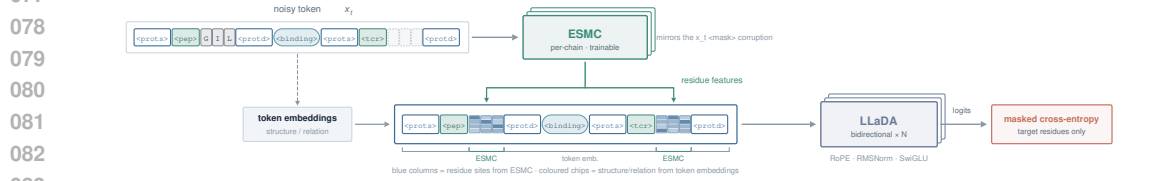
- **A unified, plug-in benchmark** (Section 2.1) covering ten downstream tasks across TCR, PPI, antibody, and nanobody families, where any embedder is selected by a single string spec.
- **Faithful baselines and calibration.** All baselines use official code and weights; reproduced baseline numbers are cross-checked against each anchor paper’s own reported values.
- **A reusable evaluation harness** (Section 2): a single frozen checkpoint is plugged into every task via post-LLaDA feature extraction. We report the integrated 7-layer checkpoint honestly, including limitations on PPI classification, unseen-epitope generalisation, and conditional generation.



(a) A single frozen checkpoint serves both understanding (prediction / representation) and generation across TCR, antibody, PPI and nanobody tasks.



(b) Designed grammar-v2 token format. Entity blocks (<prots>...<protb>) carry an optional type tab and residue span; relation tokens join blocks. Grey residues are fixed context; orange residues are diffusion targets.



(c) Training forward pass. Noisy tokens are read two ways: ESMC encodes each chain (corruption-mirrored) into residue features that replace embeddings at residue sites, while structure/relation tokens keep token embeddings; the fused stream enters a bidirectional LLaDA trained with masked cross-entropy on target residues.



(d) Inference: confidence-ordered iterative unmasking. Fixed context stays visible while masked target sites are revealed from high- to low-confidence.

Figure 1: Overview of OPHIUCHUS-X, a grammar-conditioned ESMC→LLaDA diffusion language model.

2 DOWNSTREAM BENCHMARK

We evaluate our model, OPHIUCHUS-X (a grammar-conditioned ESMC→LLaDA diffusion language model), on a unified panel of immune-receptor and protein-protein interaction downstream tasks. The row labelled OPHIUCHUS-X-integrated is the retrained integrated 7-layer checkpoint; its features are extracted *post-LLaDA* following the mandatory protocol in Section 2.1: the full record is rendered, passed through a clean (noise-free) ESMC→LLaDA forward pass, and mean-pooled once over all residue positions into a single vector. All baselines use official

Table 1: T1 TCR–epitope binding on the Nature Methods 2025 benchmark (CDR3 β , AS negatives, overall AUPRC / AUROC). Best released-model value per column in **bold**. Eight further official variants are in Table 14.

Method	Seen AUPRC	Unseen AUPRC	Seen AUROC	Unseen AUROC
ATM-TCR (official)	0.696	0.522	0.648	0.522
TEIM (official)	0.676	0.518	0.602	0.516
NetTCR-2.2 (official)	0.611	0.505	0.590	0.497
ERGO-II (official)	0.545	0.510	0.519	0.507
epiTCR (official)	0.557	0.482	0.536	0.482
PanPep (official)	0.499	0.479	0.507	0.483
CDR3-kNN (training-free)	0.553	0.544	0.507	0.504
Random	0.512	0.503	0.517	0.520
OPHIUCHUS-X-integrated	0.528	0.525	0.500	0.522

code and released weights; every baseline number is traceable to a result file under `outputs/`. Numbers for OPHIUCHUS-X-integrated come from the retrained integrated 7-layer checkpoint (`grammar_v2_esmc300m_integrated_llada_7l`, step 16,000), reported honestly, including cases where OPHIUCHUS-X trails specialised baselines.

2.1 MODEL PLUG-IN INTERFACE

Every task exposes a single string-spec entry point so that any embedder — ours or a baseline — can be swapped in without touching task code. The factory `build_embedder(spec)` routes on the prefix:

Spec prefix	Meaning
<code>bioseq-llada:/abs/best.pt</code>	our model, post-LLaDA features (headline)
<code>grammar:decoder:global:/abs/best.pt</code>	post-LLaDA whole-record global mean-pool
<code>grammar:encoder:/abs/best.pt</code>	ESMC-encoder-only ablation (control)
<code>esm2_150m, protbert, scepter, ...</code>	reference / baseline embedders

The same interface is reused by the TCR harness (`-embedder`), the MINT PPI harness (`-model grammar:.../best.pt`), and the antibody/nanobody generative scripts. Plugging a newly trained checkpoint into the full benchmark therefore requires only pointing the spec at the new `best.pt`.

2.2 TCR TASKS

T1: TCR–epitope binding (Nature Methods 2025 split). Overall (pooled) AUPRC on the official seen/unseen epitope splits (Lu et al., 2025), CDR3 β track, AS negatives (Table 1). Consistent with the anchor benchmark, all methods — including six official released models — collapse to near-random on *unseen* epitopes; OPHIUCHUS-X-integrated likewise sits near random on unseen (AUPRC 0.525) while remaining well below specialised binders on seen epitopes. We reproduce eight additional official variants (TEINet, four ERGO variants, two TPBTE variants) and the negative-source (AS/PS/HS) and near-duplicate analyses in Appendix A.4; they tell the same story.

T2: TCR clustering. We use two complementary bases. *Basis A* is the NAR Genomics & Bioinformatics 2025 paired-chain benchmark (i3 unit, 2025): we parse the authors’ official pre-computed cluster assignments for nine methods and recompute Purity/Retention/Sensitivity under their protocol (Table 2); 6/9 methods match the paper’s Purity/Retention within rounding, validating our metric implementation. *Basis A* is baseline-only (its tools do not expose an embedding, so OPHIUCHUS-X is not directly comparable). *Basis B* is the embedding→clustering protocol of TCREmbedding (Feng et al., 2025): K -means ARI/NMI/Purity averaged over $K \in \{10, \dots, 100\}$ on the 9,033 CDR3 β / 25-epitope dataset (Table 3). Rather than transcribing that paper’s unverified per-method numbers, we re-run four canonical training-free handcrafted encodings under the identical protocol: k -mer and

Table 2: T2 Basis A: NAR-GAB 2025 official nine-method clustering (paired-TCR unit, $N=5,368$). Purity–Retention trade-off recomputed from the authors’ released cluster outputs (HD/LD recomputed via string distance). Sens(top6) is sensitivity over the six highlighted pMHCs.

Method (official output)	Purity	Retention	Sens(top6)	%clu pur>0.9
clusTCR	0.991	0.079	0.135	0.964
TCRMatch	0.966	0.105	0.127	0.866
GIANA	0.931	0.175	0.290	0.714
iSMART	0.917	0.174	0.281	0.683
GLIPH2	0.904	0.186	0.277	0.638
HD / LD	0.860	0.241	0.342	0.615
DeepTCR	0.630	0.821	0.376	0.194
TCRdist3	0.421	0.387	0.004	0.070

Table 3: T2 Basis B: TCR clustering (TCREmbedding, K -means mean over K). SCEPTR uses CDR3 β +TRBV (native partial input); all other rows use CDR3 β only. Best value per column in bold.

Embedding	ARI	NMI	Purity
SCEPTR (contrastive TCR)	0.033	0.159	0.339
TCR-BERT	0.016	0.095	0.287
k -mer(2)	0.010	0.079	0.268
ESM2-150M	0.007	0.068	0.258
one-hot (composition)	0.007	0.064	0.254
ProtBERT	0.006	0.061	0.248
BLOSUM62 (mean profile)	0.004	0.056	0.240
Atchley (5-factor)	0.002	0.047	0.233
OPHIUCHUS-X-integrated	0.019	0.119	0.294

one-hot amino-acid composition, the BLOSUM62 mean substitution profile, and the Atchley five-factor descriptor. The result partially supports TCREmbedding’s finding that simple features rival data-driven PLM embeddings on CDR3 β : composition encodings (k -mer, one-hot) match or beat the general PLMs (ESM2-150M, ProtBERT) while the descriptor encodings (BLOSUM62, Atchley) trail; only the specialised TCR models (SCEPTR, TCR-BERT) lead clearly, so the decisive factor here is TCR-specific modelling rather than handcrafted vs. learned features. SCEPTR (Nagano et al., 2024) leads; OPHIUCHUS-X-integrated lands between TCR-BERT and the composition encodings, ahead of ESM2-150M/ProtBERT.

T3: TCR representation (SCEPTR few-shot, deep 6-pMHC). Per-pMHC nearest-neighbour AUROC (Nagano et al., 2024) (macro over 6 pMHC, 100 seeds) at increasing reference-set sizes k (Table 4). We reproduce the SCEPTR ordering (SCEPTR $\geq$ sequence-alignment $\geq$ general/specialised PLMs); OPHIUCHUS-X-integrated tracks ESM2-150M / TCR-BERT across k . The complementary broad 24-epitope track and the 24-way linear-probe view are in Appendix A.5.

T4: TCR generation (TCRT5 protocol (Karthikeyan et al., 2025)). Unconditional repertoire matching (Setting A, k -mer JSD $\downarrow$) and epitope-conditioned CDR3 β design (Setting B, benchmark14, $K=100$) in Table 5, the target-rich held20 split in Table 6, and full-length α/β assembly (Setting C) in Table 7. OPHIUCHUS-X-integrated matches repertoire k -mer statistics only loosely (JSD 0.295) and lags specialised conditional generators on edit distance / sequence recovery; the specialised baselines are fixed. On benchmark14 all methods score F1 $\approx$ 0 (sparse references), so seq-recovery and edit distance are the primary readouts.

2.3 PROTEIN–PROTEIN INTERACTION (MINT GENERALPPI)

We follow the MINT GeneralPPI two-stage protocol (Ullanat et al., 2026) (cache chain embeddings, train a lightweight head) with a common cap of max_train=3000 across all backends

Table 4: T3 TCR representation, deep 6-pMHC few-shot NN AUROC (100 seeds, random=0.5). Best value per column in **bold**.

Method	$k=1$	$k=5$	$k=20$	$k=100$	$k=200$
SCEPTR (contrastive)	0.640	0.702	0.738	0.775	0.790
TCRdist (official)	0.647	0.701	0.733	0.764	0.777
CDR3-Levenshtein-NN	0.626	0.671	0.696	0.722	0.733
k -mer(3)	0.607	0.665	0.690	0.717	0.728
ESM2-150M	0.591	0.631	0.656	0.685	0.696
ProtBERT	0.585	0.624	0.648	0.681	0.694
TCR-BERT	0.579	0.631	0.656	0.681	0.692
OPHIUCHUS-X-integrated	0.581	0.622	0.648	0.671	0.681

Table 5: T4 TCR generation. Left: Setting A unconditional ($n=5000$). Right: Setting B epitope-conditioned design (benchmark14, $K=100$). $\downarrow$ lower is better. OPHIUCHUS-X(unseen) is the zero-leakage epitope subset.

Setting A		Setting B (benchmark14)				
Method	JSD $\downarrow$	Method	F1@100	$d_{\text{edit}} \downarrow$	seq-rec	Char-BLEU
soNNia (official)	0.043	TCRT5 (official)	0.0013	4.31	0.617	0.716
OLGA (official)	0.061	TcrDesign-G (official)	0.0011	4.32	0.614	0.656
OPHIUCHUS-X-integrated	0.295	TCRDiff (official)	0.0009	4.72	0.587	0.527
		GRATCR (official)	0.000	4.32	0.602	0.688
		ER-Transformer (official)	0.000	5.95	0.326	0.621
		OLGA (lower bound)	0.000	5.93	0.501	0.534
		OPHIUCHUS-X-integrated	0.000	7.91	0.368	0.132
		OPHIUCHUS-X-integrated (unseen)	0.000	7.76	0.375	0.201

for apples-to-apples comparison.¹ OPHIUCHUS-X-integrated trails MINT and ESM2-650M on the classification tasks (near-random on Burnett) but is competitive on SKEMPI Spearman and leads the three-way PDB-Bind comparison under the shared cap. Four additional PLM baselines (ESM-1b, ESM2-3B, ProtT5-XL, ProGen2) and the Accuracy/F1 view are reported in Appendix A.6; ProtT5-XL is the strongest classifier on the unbiased Burnett split.

2.4 ANTIBODY TASKS

CDR infilling (SabDab 10-fold). Average amino-acid recovery (AAR, %) across folds under zero-shot masked infilling (Table 10). OPHIUCHUS-X-integrated trails AntiBERTy/AbLang2 on CDR-H1/H2 but leads on the harder CDR-H3 recovery.

Heavy $\rightarrow$ light pairing (OAS holdout500). Generated light chains scored by ImmunoMatch against the reference (Table 11). To make the comparison fair, all rows use the *same* 3-residue light prompt (prompt3); the prompt-free (de-novo) variants of p-IgGen/LICHEN and OPHIUCHUS-X are in Appendix A.7. The p-IgGen/LICHEN baselines are fixed. Note that OPHIUCHUS-X’s mask-fill decoding reveals the reference length, so its chain/V-gene match and diversity are not a like-for-like comparison with the variable-length de-novo generators (discussed in Appendix A.7).

FLAb property regression. Spearman correlation on four Graylab FLAb fitness datasets using frozen embeddings + Ridge (Table 12).

¹This shared training cap and several re-partitioned splits/data sources (e.g. the IRBench 90/90 Burnett subset, an HF-sourced PDB-Bind) mean the absolute values are *not* a byte-level reproduction of the MINT paper’s full-training numbers; they are a fair cross-backend comparison on identical data. See Appendix A.2.

Table 6: T4 Setting B on the target-rich held20 split ($K=100$). [†]For OPHIUCHUS-X this split has $\sim 48\%$ training-pair leakage, so its row is a *seen*, optimistic control only; TcrDesign-G is the strongest official conditional generator here.

Method	F1@100	$d_{\text{edit}} \downarrow$	seq-rec	Char-BLEU
TcrDesign-G (official)	0.153	1.47	0.870	0.986
TCRT5 (HF beam)	0.086	1.82	0.834	0.947
OPHIUCHUS-X-integrated [†]	0.000	6.36	0.454	0.411

Table 7: T4 Setting C, full-length α/β assembly. OPHIUCHUS-X (unconditional, $n=500$) attains high per-residue AA validity but only a small fraction of β chains yield an ANARCI-extractable canonical CDR3 β — its honest capability boundary is CDR3-level, not full-length de novo. TcrDesign (epitope-conditioned, 34 pMHC $K=100$) is the full-length SOTA. “—” = not applicable to that mode.

Method	cond.	valid-AA (β/α)	CDR3 extract	F1@100	seq-rec
OPHIUCHUS-X-integrated (full-length)	uncond.	0.998 / 0.996	0.038	—	—
TcrDesign (full-length)	epitope	—	—	0.061	0.772

2.5 NANOBODY MULTI-TASK (NB BENCH)

Post-LLaDA probing across the 12 NbBench tasks (Table 13). We report four comparators: our re-run ESM2-150M (identical frozen-backbone + probe pipeline as OPHIUCHUS-X, the fair same-pipeline reference), the official ESM2-150M (transcribed from the NbBench paper), the best official model per task (of 11), and OPHIUCHUS-X. The three reference columns are fixed. The official columns use the NbBench MLP-head pipeline and are shown for context (full 11-model leaderboard in Appendix A.8); the thermo tasks carry a near-duplicate caveat also noted there. OPHIUCHUS-X-integrated is competitive on classification/regression probes and leads the same-pipeline hTNFa AUROC.

2.6 SUMMARY

The panel spans 10 downstream tasks with fixed, paper-calibrated external baselines and a single plug-in interface. Under the integrated 7-layer checkpoint, OPHIUCHUS-X is competitive on antibody CDR-H3 recovery, MINT affinity regression (SKEMPI / PDB-Bind), and several NbBench probes, while trailing specialised baselines on TCR binding/clustering/few-shot, unconditional repertoire matching, PPI classification, and epitope-conditioned TCR design — reported without inflation.

Table 8: MINT GeneralPPI binary classification (test set, 3-rep mean). HumanPPI/Bernett/YeastPPI report AUROC/AUPRC; MutationalPPI is class-imbalanced (read AUPRC). Best per row in **bold**.

Task	Metric	MINT	ESM2-650M	OPHIUCHUS-X-integrated
HumanPPI	AUROC	0.938	0.939	0.688
	AUPRC	0.910	0.941	0.720
Bernett	AUROC	0.739	0.786	0.519
	AUPRC	0.748	0.767	0.523
YeastPPI	AUROC	0.678	0.648	0.567
	AUPRC	0.717	0.674	0.583
MutationalPPI	AUPRC	0.186	0.250	0.176

Table 9: MINT GeneralPPI regression. SKEMPI $\Delta\Delta G$ (3-fold complex-held-out) and PDB-Bind affinity (10-fold Ridge CV). Higher Spearman / lower RMSE is better. **bold** marks the best value per column.

Backend	SKEMPI		PDB-Bind	
	Spearman	RMSE↓	Spearman	RMSE↓
MINT (official)	0.208	5.92	0.615	0.949
ESM2-650M	0.251	3.69	0.570	1.094
OPHIUCHUS-X-integrated	0.308	1.01	0.641	0.767

Table 10: Antibody CDR infilling, average AAR (%) over SabDab 10-fold. Best per column in **bold**. Baselines are re-run with official weights under zero-shot masked infilling; the Ophiuchus-Ab paper’s own Table 2 uses a different masking protocol and is not transcribed here.

Model	CDR-H1	CDR-H2	CDR-H3
AntiBERTy (zero-shot)	76.20	71.61	41.23
AbLang2 (zero-shot)	76.29	70.69	41.66
OPHIUCHUS-X-integrated	72.27	67.78	48.79

Table 11: Antibody heavy→light pairing (OAS holdout500, $n=8$ /heavy, prompt3 for all rows). Reference ImmunoMatch =0.699; the Ophiuchus-Ab paper reports ≈ 0.701 under this protocol. Best per column in **bold**. [‡]Chain match for OPHIUCHUS-X is inflated by mask-fill length revelation (see text); not ranked against de-novo generators.

Model	ImmunoMatch	gen>ref	chain match	diversity
p-IgGen (prompt3)	0.683	0.473	0.990	0.368
LICHEN (prompt3)	0.632	0.421	0.998	0.255
OPHIUCHUS-X-integrated	0.600	0.364	1.000 [‡]	0.118

Table 12: FLAb antibody property regression (Spearman, 5-fold CV). Best per column in **bold**. This is an independent sanity-check, not a reproduction of the Ophiuchus-Ab GDPa1 developability table.

Model	g6 Kd	g6 ER	trastuzumab	d44	mean
One-hot	0.538	0.602	0.552	0.473	0.541
ProtBERT	0.579	0.766	0.545	0.539	0.607
ESM-2 650M	0.611	0.820	0.586	0.569	0.647
Ophiuchus-Ab	0.616	0.795	0.612	0.556	0.645
IgBERT	0.559	0.732	0.599	0.589	0.620
AntiBERTy	0.488	0.666	0.606	0.504	0.566
OPHIUCHUS-X-integrated	0.569	0.742	0.548	0.453	0.578

Table 13: NbBench 12-task probing. Metrics: accuracy (VRClassification, nanobody_type), BLOSUM62 recovery (CDRInfilling), AUPRC (Paratope), AUROC (SARS-CoV-2/hIL6/hTNFa/polyreaction), Spearman (thermo-tm/-seq, vhh_affinity-seq/-score). “E150 (ours)” and OPHIUCHUS-X share our frozen-probe pipeline; “E150 (off.)” and “Best off.” are transcribed from NbBench. Best-official codes: AbH=AbLang-H, E650=ESM2-650M, Ab2C=AntiBERTa2-CSSP, Ab2=AntiBERTa2, AbTy=AntiBERTy, IgB=IgBert, AbL=AbLang-L. Best per row across all columns in **bold**.

Task	Metric	E150 (ours)	E150 (off.)	Best off.	OPHIUCHUS-X
VRClassification	Acc	0.9989	0.9989	0.9989	0.9785
nanobody_type	Acc	0.9997	0.994	0.999 (AbTy)	0.9952
CDRInfilling	BR	1.467	1.499	1.551 (E650)	1.437
Paratope	AUPRC	0.682	0.586	0.765 (Ab2C)	0.640
SARS-CoV-2	AUROC	0.868	0.834	0.884 (AbH)	0.814
hIL6	AUROC	0.898	0.850	0.925 (AbH)	0.767
polyreaction	AUROC	0.865	0.833	0.842 (E650)	0.852
thermo-tm	Spearman	0.658	0.301	0.593 (Ab2C)	0.570
thermo-seq	Spearman	0.625	0.389	0.587 (Ab2)	0.597
vhv_affinity-seq	Spearman	0.195	0.170	0.184 (IgB)	0.181
vhv_affinity-score	Spearman	0.165	0.063	0.128 (AbL)	0.143
hTNFa	AUROC	0.788	–	–	0.843

3 CONCLUSION

We assembled a unified, plug-in downstream benchmark for an immune-receptor and PPI sequence foundation model, spanning ten tasks with official baselines and paper-calibrated reproductions. On the retrained integrated 7-layer checkpoint, OPHIUCHUS-X is competitive on antibody CDR-H3 recovery, MINT affinity regression (SKEMPI / PDB-Bind), and several NbBench probes, while trailing specialised baselines on TCR binding, clustering, few-shot representation, unconditional repertoire matching, PPI classification, and epitope-conditioned TCR design. The string-spec interface makes the panel directly reusable for any future checkpoint. Future work includes running the MINT tasks under the paper’s full (uncapped) training protocol for a byte-level reproduction, and extending the antibody suite to developability once GDPa1 data are available.

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A REPRODUCIBILITY AND CAVEATS

A.1 CHECKPOINTS

All headline OPHIUCHUS-X-integrated numbers in the main text use the retrained integrated 7-layer checkpoint `grammar_v2_esmc300m_integrated_llada_7l` (best.pt at step 16,000; LLaDA `num_hidden_layers=7`, `hidden_size=512`), an 8-source mixture trained from scratch with training shards deduplicated against every downstream test set. Features are extracted post-LLaDA via `grammar:decoder:global:.` Earlier `cmp500k/mint/28-layer` integrated checkpoints are retained on disk for ablation but are not the headline rows. Appendix tables that still show *tbd* (T1 near-dedup enc-only control, T3 broad/probe, de-novo pairing) were not re-run for this checkpoint.

A.2 MINT GENERALPPI PROTOCOL CAVEAT

All MINT backends share a training cap of `max_train=3000` for a fair cross-backend comparison, and several tasks use re-partitioned splits or alternative data sources: the Bennett task uses the IRBench 90/90 subset (test =2644) rather than the paper’s full gold-standard test (=52048), and PDB-Bind is built from the HuggingFace `proteinea/ppb_affinity` mirror. Consequently the absolute values are *not* a byte-level reproduction of the MINT paper (which trains on the full data and reports AUPRC =0.69 on the full Bennett test); under our shared cap the relative ranking on classification differs from the paper (e.g. ESM2-650M / ProtT5-XL can exceed MINT). All three backends read identical CSVs, so the comparison is internally fair. A faithful full-training reproduction is left to future work.

A.3 DATA PROVENANCE

Every number in Section 2 is traceable to a result file under `outputs/` (TCR/PPI/NbBench) or `output/downstream_generation/` (antibody). Baseline reproductions were calibrated against the anchor papers: the TCR binding, clustering, and representation baselines match the Nature Methods 2025, NAR-GAB 2025, and SCEPTR reference values within the tolerances reported in the respective source repos; the TCR generation metric implementation matches the official TCRT5 evaluator (42 prediction blocks, zero mismatches); and the NbBench probes match the official ESM2-150M reference.

A.4 T1: ADDITIONAL OFFICIAL BASELINES AND ANALYSES

Eight further official variants. Beyond the six released models in Table 1, we reproduce eight additional official variants from the Nature Methods 2025 repository under the identical cdr3b/AS protocol (Table 14). All show the same pattern: usable signal on seen epitopes (AUPRC 0.50–0.67) collapsing to near-random on unseen (0.50–0.52), reinforcing that unseen-epitope generalisation failure is field-wide.

Table 14: T1 additional official baselines (cdr3b, AS, overall AUPRC/AUROC).

Method	Seen AUPRC	Unseen AUPRC	Seen AUROC	Unseen AUROC
TEINet-large	0.666	0.505	0.601	0.502
TEINet-small	0.656	0.501	0.592	0.505
ERGO-LSTM (VDJdb)	0.628	0.519	0.577	0.512
ERGO-AE (VDJdb)	0.615	0.507	0.543	0.499
ERGO-LSTM (McPAS)	0.609	0.509	0.561	0.511
ERGO-AE (McPAS)	0.580	0.523	0.515	0.505
TPBTE (McPAS)	0.501	0.501	0.505	0.504
TPBTE (VDJdb)	0.500	0.500	0.500	0.500

Negative-source sensitivity. Swapping the negative-sampling source (AS/PS/HS) moves overall AUPRC by as much as the gaps between models (Table 15), reproducing the anchor paper’s finding that negative-set construction can matter more than model architecture. HS systematically inflates scores (mixed-source spurious signal), so the main table fixes AS.

Table 15: T1 negative-source sensitivity (overall AUPRC). AS/PS/HS are the three official negative-sampling schemes.

Method	Seen			Unseen		
	AS	PS	HS	AS	PS	HS
ATM-TCR	0.696	0.682	0.666	0.522	0.508	0.519
NetTCR-2.2 (b)	0.611	0.622	0.578	0.505	0.572	0.526
TEIM	0.676	0.700	0.741	0.518	0.536	0.596
ERGO-II	0.545	0.556	0.554	0.510	0.507	0.509
epiTCR (noMHC)	0.557	0.541	0.543	0.482	0.500	0.490
PanPep	0.499	0.504	0.590	0.479	0.448	0.510
CDR3-kNN ($k=5$)	0.553	0.526	0.649	0.544	0.544	0.544
OPHIUCHUS-X-integrated	0.528	0.517	0.622	0.525	0.480	0.559

Near-duplicate removal. Removing test rows whose CDR3 β lies within Levenshtein $\leq k$ of any training CDR3 β (Table 16) shows that (i) unseen AUPRC stays near 0.5 for all methods (the generalisation failure is real, not a duplicate artifact), while (ii) the training-free CDR3-kNN’s apparent unseen edge collapses as soon as near-duplicates are removed ($0.544 \rightarrow 0.487$ at $k \leq 1$).

Table 16: T1 near-duplicate removal (unseen split, overall AUPRC). Columns are increasing dedup radius k ; “kept” is the number of remaining unseen test rows.

	$k=0$	$k \leq 1$	$k \leq 2$	$k \leq 3$
unseen rows kept	690	559	225	51
CDR3-kNN	0.544	0.487	0.499	0.490
OPHIUCHUS-X-integrated (enc-only)	<i>tbd</i>	<i>tbd</i>	<i>tbd</i>	<i>tbd</i>
ATM-TCR	0.522	0.523	0.564	0.542
TEIM	0.518	0.507	0.528	0.560

A.5 T3: BROAD-TRACK AND LINEAR-PROBE VIEWS

The complementary broad 24-epitope few-shot track (clonotype-disjoint split, Table 17) reproduces the deep-track ordering on a larger epitope surface. The 24-way linear-probe view (Table 18) has ample supervision so most embeddings read out epitope identity in a narrow band (0.75–0.825 probe AUROC), which is why the few-shot protocol is our primary representation metric; SCEPTR’s kNN top-1 lead (0.506) reflects its more epitope-clustered representation.

Table 17: T3 broad 24-epitope few-shot NN AUROC (random=0.5).

Method	$k=1$	$k=5$	$k=20$	$k=100$
SCEPTR (contrastive)	0.587	0.655	0.711	0.741
TCRdist (official)	0.577	0.644	0.687	0.728
CDR3-Levenshtein-NN	0.555	0.597	0.641	0.683
k -mer(3)	0.543	0.597	0.643	0.673
ESM2-150M	0.561	0.589	0.638	0.671
ProtBERT	0.550	0.587	0.627	0.651
TCR-BERT	0.547	0.577	0.623	0.641
Ophiuchus-Ab	0.576	0.612	0.664	0.677
OPHIUCHUS-X-integrated	<i>tbd</i>	<i>tbd</i>	<i>tbd</i>	<i>tbd</i>

Table 18: T3 24-way linear-probe / 1-NN (backward-compatible auxiliary view).

Method	probe-AUROC	probe-Acc	kNN top-1
ESM2-150M	0.806	0.438	0.409
ProtBERT	0.793	0.405	0.393
TCR-BERT	0.748	0.416	0.431
Ophiuchus-Ab	0.825	0.452	0.443
k -mer(3)	0.800	0.430	0.447
SCEPTR (64-d)	0.803	0.425	0.506
OPHIUCHUS-X-integrated	<i>tbd</i>	<i>tbd</i>	<i>tbd</i>

A.6 MINT: ADDITIONAL PLM BASELINES AND ACCURACY/F1

Under the shared `max_train=3000` cap, four further PLM backbones on the two classification tasks with a released comparison (Table 19) confirm the MINT paper’s trend: ProtT5-XL is strongest on the unbiased Bennett split (AUROC 0.799, ahead of MINT-official 0.739), while the autoregressive ProGen2 is weakest. Table 20 gives the Accuracy/F1 view underlying Table 8.

A.7 ANTIBODY PAIRING: PROMPT-FREE (DE-NOVO) VARIANTS

Table 11 uses `prompt3` for all rows. Table 21 gives the prompt-free (de-novo) numbers; the OPHIUCHUS-X-integrated de-novo row was not re-run for the 7-layer checkpoint (main comparison uses `prompt3`). A key honesty note that still applies: OPHIUCHUS-X’s mask-fill decoding uses a number of mask tokens equal to the reference light-chain length, so even in de-novo mode it

Table 19: MINT classification, additional PLM baselines (AUROC/AUPRC, 3-rep mean, -sep_embed). Only HumanPPI and Bennett have a released comparison.

Backbone	HumanPPI		Bennett	
	AUROC	AUPRC	AUROC	AUPRC
ESM-1b	0.937	0.931	0.727	0.711
ESM2-3B	0.938	0.938	0.711	0.694
ProtT5-XL	0.939	0.941	0.799	0.777
ProGen2-base	0.925	0.906	0.622	0.603

Table 20: MINT classification, Accuracy/F1 (3-rep mean).

Task	Metric	MINT	ESM2-650M	OPHIUCHUS-X-integrated
HumanPPI	Acc	0.876	0.851	0.646
	F1	0.874	0.855	0.615
Bennett	Acc	0.661	0.709	0.509
	F1	0.625	0.686	0.314

reconstructs the target length, yielding chain-match ≈ 1.0 and low diversity by construction — so its match/diversity columns are *not* a like-for-like advantage over the truly variable-length de-novo generators p-IgGen/LICHEN.

Table 21: Antibody heavy→light pairing, de-novo (prompt-free) variants (OAS holdout500, $n=8$ /heavy). Reference ImmunoMatch =0.699.

Model	ImmunoMatch	gen>ref	chain match	V (exact)	diversity
p-IgGen	0.686	0.482	0.552	0.050	0.695
LICHEN	0.666	0.498	0.622	0.053	0.519
OPHIUCHUS-X-integrated	<i>tbd</i>	<i>tbd</i>	<i>tbd</i>	<i>tbd</i>	<i>tbd</i>

A.8 NbBENCH: FULL 11-MODEL OFFICIAL LEADERBOARD

Tables 22 and 23 transcribe the complete NbBench 11-model leaderboard (from the anchor paper, official MLP-head pipeline) and append our same-pipeline runs (frozen backbone + probe): our re-run ESM2-150M and OPHIUCHUS-X-integrated. Paratope is omitted here because the official leaderboard reports AUROC while our pipeline reports AUPRC (both given in Table 13). **Caveat:** the thermo tasks have substantial train/test near-duplication ($\sim 52\%$ / $\sim 32\%$ of test within edit distance 2 of train under NbBench’s 75%-similarity split), so composition- sensitive models are inflated there; the affinity tasks are the cleaner regression signal.

Table 22: NbBench full leaderboard, group A. VRCIs/nanobody_type = Accuracy, CDRInfilling = BLOSUM62 recovery, SARS-CoV-2/hIL6/polyreaction = AUROC. Rows above the rule are transcribed from NbBench; rows below use our pipeline.

Model	VRCIs	nb_type	CDRInf	SARS	hIL6	poly
ProtBERT	0.998	0.957	1.520	0.852	0.878	0.837
ESM2-150M	0.999	0.994	1.499	0.834	0.850	0.833
ESM2-650M	0.998	0.995	1.551	0.850	0.857	0.842
AbLang-H	0.998	0.997	1.452	0.884	0.925	0.831
AbLang-L	0.998	0.988	1.360	0.811	0.914	0.819
AntiBERTy	0.998	0.999	1.499	0.810	0.871	0.828
AntiBERTa2	0.999	0.995	1.489	0.824	0.887	0.833
AntiBERTa2-CSSP	0.998	0.998	1.484	0.845	0.916	0.830
IgBert	0.998	0.993	1.311	0.817	0.821	0.829
NanoBERT	0.995	0.988	1.524	0.876	0.904	0.815
VHHBERT	0.990	0.986	1.244	0.800	0.873	0.818
ESM2-150M (ours pipe.)	0.999	1.000	1.467	0.868	0.898	0.865
OPHIUCHUS-X-integrated	0.978	0.995	1.437	0.814	0.767	0.852

Table 23: NbBench full leaderboard, group B (Spearman). Rows above the rule are transcribed from NbBench; rows below use our pipeline.

Model	thermo-tm	thermo-seq	vhh_aff-seq	vhh_aff-score
ProtBERT	0.128	0.040	0.163	0.084
ESM2-150M	0.301	0.389	0.170	0.063
ESM2-650M	0.315	0.375	0.183	0.089
AbLang-H	0.533	0.566	0.151	0.106
AbLang-L	0.502	0.585	0.173	0.128
AntiBERTy	0.385	0.474	0.167	0.119
AntiBERTa2	0.585	0.587	0.162	0.088
AntiBERTa2-CSSP	0.593	0.575	0.149	0.085
IgBert	0.344	0.351	0.184	0.090
NanoBERT	0.298	0.440	0.061	0.118
VHHBERT	0.532	0.484	0.145	0.115
ESM2-150M (ours pipe.)	0.658	0.625	0.195	0.165
OPHIUCHUS-X-integrated	0.570	0.597	0.181	0.143